Functions and Recursion Practice Sheet

Easy

1. Function Basics

Write a simple function that prints a greeting message.

2. Sum of Two Numbers

Write a function that takes two numbers as arguments and returns their sum.

3. Check Even or Odd

Write a function that checks whether a number is even or odd.

4. Find Maximum

Write a function that returns the maximum of three numbers.

5. Factorial using Loop

Calculate factorial of a number using an iterative function.

6. Prime Check

Write a function to check if a number is prime.

7. Function Overloading (C++)

Demonstrate function overloading with different parameter sets (C++ only).

Medium

1. Pass by Value vs Pass by Reference

Demonstrate difference between pass by value and pass by reference using a simple swap function.

2. Power Function

Write a function to compute a^b using a loop.

3. Count Digits

Write a function that returns the number of digits in a given number.

4. Sum of Digits

Write a function to calculate the sum of digits of a number.

5. Reverse a Number

Write a function to reverse a number using logic and return the reversed number.

6. Check Armstrong Number

Write a function that checks if a number is an Armstrong number.

7. GCD using Euclidean Algorithm

Write a function that returns the GCD of two numbers using the Euclidean algorithm.

Hard (Recursion)

1. Factorial using Recursion

Write a recursive function to compute the factorial of a number.

2. Fibonacci using Recursion

Write a recursive function to print the nth Fibonacci number.

3. Sum of Natural Numbers

Write a recursive function that returns the sum of first n natural numbers.

4. Power Function using Recursion

Write a recursive function to calculate a^b.

5. Check Palindrome (Recursion)

Write a recursive function to check if a string is a palindrome.

6. Decimal to Binary

Write a recursive function to convert a decimal number to binary.

7. Tower of Hanoi

Solve the Tower of Hanoi problem using recursion for n disks.

8. Reverse a String (Recursion)

Write a recursive function to reverse a string.

9. Generate All Subsets

Write a recursive function to generate all subsets of a string.

10. N-th Stair Problem

You can climb 1 or 2 steps. Count the ways to reach the nth step using recursion.